

# ASTRAL Supplementary Material

This document provides supplementary information for our work on ASTRAL [3]. It includes detailed derivations, additional experimental results, implementation details, and extended analyses that complement the main paper.

## 1 Probabilistic Analysis Formulations

This section summarises the probabilistic analysis formulations used in ASTRAL, adapted from our earlier work on Bayesian and multi-objective decision support [2].

### 1.1 Probabilities of Exposure

For vulnerability nodes, a distinction is made between those with public disclosure via Common Vulnerabilities and Exposures (CVE) identifiers and those that are proprietary or undocumented, lacking formal CVE enumeration. Vulnerabilities with CVE identifiers are assessed using the Common Vulnerability Scoring System (CVSS), while those without CVEs are assigned proxy-CVSS estimates based on observed characteristics. Exposure probabilities are derived by quantifying the exploitability factors defined in the CVSS specification [1], as detailed in the following formula:

$$P(Exposure)_{\text{vuln}} = AV \times AC \times PR \times UI \quad (1)$$

In addition, we introduce an attack feasibility (AF) modifier, inspired by Xie et al. [6], to further refine vulnerability exposure probability estimations and account for real-world conditions. The AF modifier adjusts the probability based on system security posture, adversary capabilities, and environmental constraints. Systems with up-to-date defences and restricted access are associated with lower AF values, reflecting reduced likelihood of successful attack. In contrast, an unpatched CPS exposed to a skilled adversary will exhibit higher AF values and consequently higher probabilities of successful attack. The final probability of successful attack for a vulnerability node is therefore given by:

$$P(Attack)_{\text{vuln}} = P(Exposure)_{\text{vuln}} \times \phi \quad (2)$$

where  $P(Attack)_{\text{vuln}}$  represents the probability of successful attack,  $P(Exposure)_{\text{vuln}}$  is the vulnerability exposure probability, and  $\phi$  is the attack feasibility modifier.

For *asset nodes*, the probability of exposure equates to probability of failure, modelled by exponential decay [4]:

$$P(Exposure)_{\text{asset}} = 1 - e^{-\lambda t} \quad (3)$$

where  $\lambda$  denotes the failure rate per unit time, and  $t$  denotes the asset's operational duration.

*Hazard nodes* represent hazardous events, failures, or disruptions influenced by attack propagation. As such, the occurrence of a hazard is influenced by the cyber-physical dependencies within the system. If the parent node of a hazard is an asset or hazard, then the failure of the parent node increases the likelihood of hazard activation and propagates risk, potentially escalating the attack scenario. The probability of hazard exposure is thus represented as:

$$P(Exposure)_{\text{hazard}} = \begin{cases} 1, & \text{if parent fails or occurs} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

## 1.2 Probabilities of Severe Impact

For *vulnerability nodes*, the probability of severe impact is computed as:

$$P(Impact)_{\text{vuln}} = 1 - [(1 - C) \cdot (1 - I) \cdot (1 - A)] \quad (5)$$

where:  $P(Impact)_{\text{vuln}}$  denotes the probability of severe impact, and  $C, I, A$  represent the CVSS impact metrics for confidentiality, integrity, and availability, respectively. For proprietary or undocumented vulnerabilities, these metrics are estimated using system context or analogous vulnerabilities.

For *asset* and *hazard nodes*, the probability of severe impact is proportional to node connectivity:

$$P(Impact)_{\text{asset, hazard}} = \frac{\text{Number of connected child nodes}}{\text{Total number of nodes in the BN}} \quad (6)$$

## 1.3 Computing Risk and Availability Scores

Once exposure and impact probabilities for vulnerability, asset, and hazard nodes are determined, the posterior probabilities of exposure (or attack success in the case of vulnerability nodes) and severe impact are calculated using conditional probability formulations and VE techniques. The composite risk score, representing the joint likelihood of successful exposure and severe impact, is computed as the product of these posterior probabilities:

$$Risk = P(Exposure)_{\text{posterior}} \times P(Impact)_{\text{posterior}} \quad (7)$$

In parallel, system availability can be derived as an inverse measure reflecting the probability that neither exposure nor severe impact occurs. This probabilistic complement captures the expected operational continuity of the system under uncertainty and is expressed as:

$$Availability = (1 - P(Exposure)_{\text{posterior}}) \times (1 - P(Impact)_{\text{posterior}}) \times 100 \quad (8)$$

In this work's context, the availability value (expressed as a percentage) quantifies the confidence that the system remains unaffected by both exploit attempts and consequential disruptions, complementing the risk score within the same probabilistic assessment framework.

## 1.4 Construction of Conditional Probability Tables

The conditional probability tables (CPTs) for BN nodes with parents are computed by multiplying parent node states with node-specific base probabilities from Equations (1)–(6). For example, consider vulnerability node  $V$  with parent asset node  $A$ . The CPT is constructed as:

$$P(V = 1|A = 0) = 0.05, \quad P(V = 1|A = 1) = P(Exposure)_{\text{vuln}} = 0.47$$

Resulting CPT:

| $A$ | $V = 0$ | $V = 1$ |
|-----|---------|---------|
| 0   | 0.95    | 0.05    |
| 1   | 0.53    | 0.47    |

## 2 Reference-Architecture Validation

The architectural artefacts used for reference-architecture validation of ASTRAL’s reconstruction correctness are reproduced here for reference.

The SWaT Network Architecture, i.e., the cyber portion of SWaT, comprises dedicated OT hardware in Allen-Bradley Programmable Logic Controllers (PLCs), Schneider Remote Input/Output (RIO) units and a combination of IT software and hardware in the Human Machine Interfaces (HMIs), Supervisory Control and Data Acquisition (SCADA) workstation and Historian.

The SCADA supervises the SWaT process through periodical monitoring and has the ability to take manual control of the SWaT testbed by overriding the pre-existing PLC programming. The SCADA also works in conjunction with the Historian which serves the important role of recording and storing process data for subsequent analysis. The IT and OT components co-exist on a network that employs a layered communications network using the Purdue Model for Industrial Control Systems (ICS). The PLCs use the EtherNet/IP (ENIP) as the main communication protocol.

Notable aspects of the testbed include segmented communications networks, both wired and wireless communications, distributed dynamic control, interconnection among the testbeds, and complete access to the control logic within the PLCs and HMIs. The SWaT testbed employs the use of Wireless Access Points (WAP) to achieve wireless communications. This accessibility allows researchers to develop and upload their code in the controllers for research and experimentation. It also provides a platform for demonstrating their technologies in a safe, controlled, and realistic environment.

Figure 1: Textual architectural artefact for Secure Water Treatment (SWaT) Testbed [5]

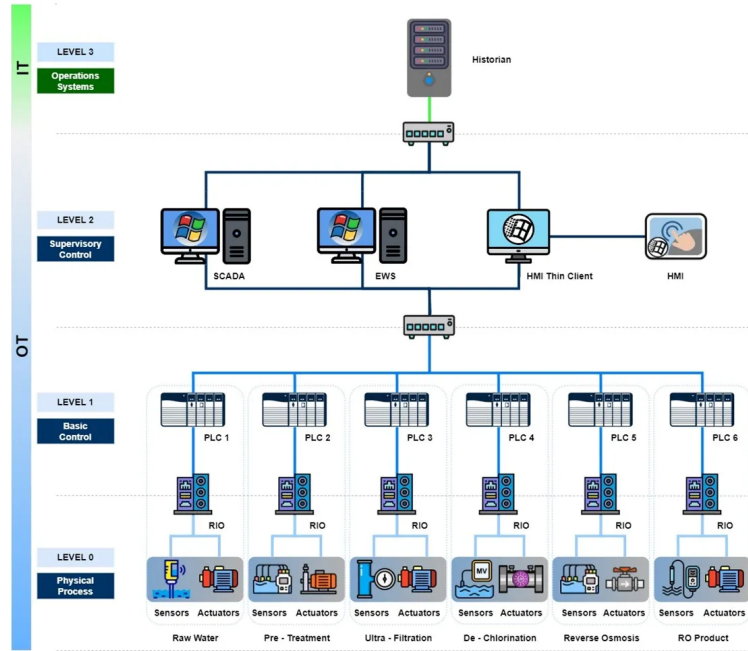


Figure 2: Reference network architecture for Secure Water Treatment (SWaT) Testbed [5]

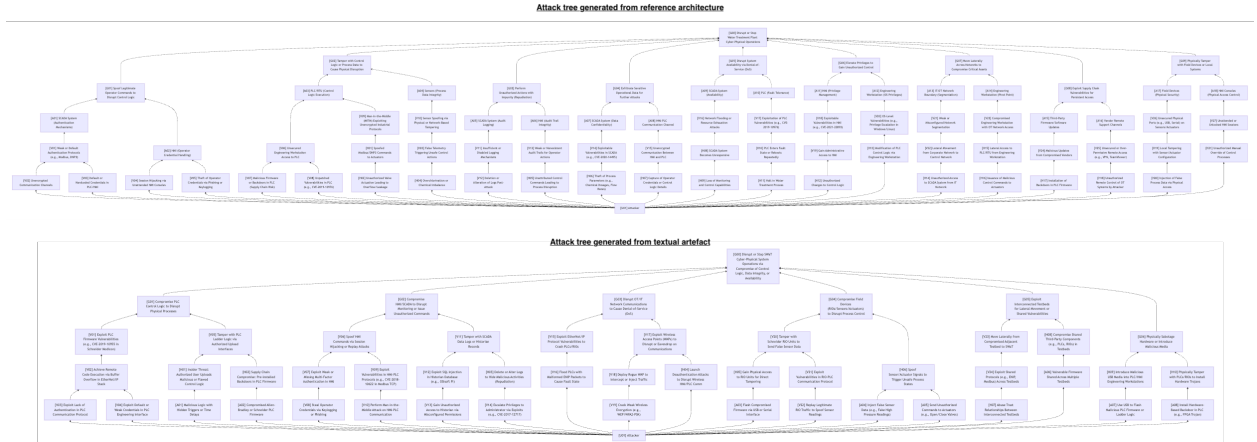


Figure 3: SWaT attack tree comparison generated by Mistral Medium 3.1 from the reference network architecture (top) and the textual architectural artefact (bottom).

Table 1: Probabilistic risk analysis comparison for the SWaT reference architecture and textual artefact across LLM configurations

| Model              | Reference Architecture | Textual Description | Variance |
|--------------------|------------------------|---------------------|----------|
| Mistral Medium 3.1 | 55.83%                 | 61.63%              | 10.39%   |
| Gemini 3.5 Flash   | 1.00%                  | 1.00%               | 0.00%    |

### 3 Ablation Study

This section presents the ablation study using three case studies: *FrostyGoop* (incomplete DFD), *Medical CPS (MCPS)* (high-level schematic), and *Solar PV Inverter* (BN graph). Four variants systematically disable or modify key design elements relative to a *Full ASTRAL* baseline (multimodal input with guardrails enabled, temperature = 0.1, top- $p$  = 0.9):

- *Text-Only* (no diagrams, guardrails on)
- *No-Guardrails* (full multimodal, guardrails off)
- *High temperature* (multimodal + guardrails, temperature = 0.9)
- *Low top- $p$*  (multimodal + guardrails, top- $p$  = 0.1)

Each variant was run ten times per case study using identical prompts and the same LLM (Mistral 3.1 Medium). The artefacts for the Text-Only experiments, the prompts for the No-Guardrails experiments, and the evaluation metrics and corresponding statistical analysis are described in the subsequent sections.

### 3.1 Artefacts for Text-Only Experiments

The artefacts used for the Text-Only experiments are as follows:

The image shows the connectivity of a utility management and automated heating system. The components are arranged in a top-down hierarchy, starting from the internet and moving down to local meters and substation modules.

#### System Connectivity and Flow

- The MicroTik Router sits at the top of the hierarchy, connected to the Internet. It has two primary downward connections: one via Ethernet to the ENCO Reader/Workstation and another via Ethernet/USB to the ENCO Controller.
- The ENCO Reader/Workstation acts as a central hub, maintaining a bi-directional Ethernet/USB link with the ENCO Controller and a USB connection to the ENCO Terminal.
- The ENCO Controller manages several data streams. It connects to the ENCO Box using RF (Radio Frequency) and to the Gas/Electric Meter using the MBus protocol.
- The ENCO Box serves as a collection point for multiple meters. It receives data from the Gas/Electric Meter via MBus and from the Water Meter via RF. It then sends data to the Heat Meter via an Ethernet connection.
- At the bottom of the diagram, the Automated Heating Substation Module receives signals from the ENCO Terminal via RF. This module then directs data or control signals via Ethernet to both the Water Meter and the Heat Allocator. Finally, the Heat Allocator is linked to the Heat Meter through an Ethernet connection.

#### Key Protocols Identified

- Ethernet/USB: Used for high-level communication between the router, workstation, and controller.
- RF (Radio Frequency): Used for wireless communication between the terminal, substation, and meters.
- MBus: A specialized bus protocol used specifically for the Gas and Electric meters to communicate with the controller and box.

Figure 4: Text-Only artefact for FrostyGoop

This diagram illustrates a Medical Cyber-Physical System (MCPS) network, showing the interconnectedness between a central medical association, hospital facilities, mobile units, and patient-specific medical devices.

- **Network Core (The Medical Association):** At the center of the architecture is the Medical Association, which acts as a hub for multiple hospital facilities. These facilities are interconnected via Network transmission equipment (represented by routers) and wireless lightning-bolt icons, indicating a high-level wide-area network.
- **Hospital Facilities and Diagnostic Equipment:** Each hospital node is connected to various heavy diagnostic and specialized systems:
  - Hyperbaric oxygen storage equipment
  - Chest X-ray equipment
  - Nuclear magnetic resonance (MRI) equipment
  - Ultrasonic diagnosis instrument
- **Mobile and Remote Communication:** The system extends beyond the physical walls of the hospital through several communication channels:
  - Remote communication: Linked to the hospital nodes to facilitate off-site data exchange.
  - Ambulance and Mobile communication: Ambulances are shown connected to the hospital network via wireless links, ensuring real-time data flow during patient transport.
- **Patient-Centric Devices and Wearables:** The bottom and left side of the diagram detail the "Cyber-Physical" aspect, focusing on devices directly interacting with the patient:
  - Implantable Devices: This includes a Brain chip and a Heart pacemaker.
  - Wearables: A Medical Bracelet is used for continuous monitoring.
- **Mobility and Rehabilitation:** The network includes a Smart Wheelchair, Leg rehabilitation equipment, and a Smart hospital bed, all of which are integrated into the hospital's data environment.

#### **Summary of System Flow**

The architecture demonstrates a nested hierarchy where:

- Personal/Physical devices feed data into local hospital systems.
- Local systems coordinate with specialized diagnostic equipment and mobile units.
- Hospital nodes synchronize at the top level under the Medical Association umbrella to ensure a unified health record and response system.

Figure 5: Text-Only artefact for MCPS

=This diagram illustrates a threat model for a Solar PV Inverter system, mapping out vulnerabilities (labeled V), hazards or impacts (labeled H), and the flow of potential cyberattacks from user-facing applications down to the physical power grid.

**System Entry Points and Application Layer** The top of the diagram identifies the primary interfaces for user interaction:

- Mobile/Web Application: Connects to several vulnerabilities including decrypting communications (V1), man-in-the-middle server impersonation (V2), and hardcoded MQTT credentials (V3).
- WiNet Web UI: Directly linked to a vulnerability involving hardcoded passwords (V4).

**Central Communication and Cloud Services** The core of the digital infrastructure involves the SunGrow iSolarCloud (MQTT Broker). This component is susceptible to multiple vulnerabilities:

- IDOR (Insecure Direct Object Reference) Vulnerabilities: A series of five distinct IDOR flaws (V6 through V10), each associated with a specific CVE from 2024, can be exploited through the cloud broker.
- Impersonation: The MQTT broker can be impersonated (V5) following an exploit of hardcoded passwords.
- Communication Hardware and Firmware Vulnerabilities: Data flows from the cloud and management layers to the WiNet-S Communication Dongle. This hardware is vulnerable to several Denial of Service (DoS) or Remote Code Execution (RCE) attacks:
  - Buffer Overflows: Three stack-based buffer overflows (V11, V12, V13) and one heap-based buffer overflow (V14).
  - Firmware Injection: An attacker can inject controlled firmware (V15) to gain control over the device.

**Cyber-Physical Impact and Grid Hazards** The bottom of the diagram shows how digital exploits translate into physical consequences for the Solar PV Inverter and the Power Grid:

- Immediate Hazards: Potential impacts include data leakage (H1), hijacking other devices (H2), and cyber-physical ransomware (H3).
- Operational Interference: Exploits can lead to altered energy production (H4) or a dynamic load attack (V16).
- Grid Consequences: These operational failures can result in load shedding (H5), emergency equipment failure (H6), and grid instability (H7).
- Ultimate Outcome: The final catastrophic impact at the bottom of the chain is a Power Outage (H8).

Figure 6: Text-Only artefact for solar PV inverter

### 3.2 LLM Prompts for No-Guardrails Experiments

The LLM prompts used for the No-Guardrails experiments are as follows:



You are a Senior Solution Architect tasked with narrating a system architectural diagram (e.g., Data Flow Diagram) to a Senior Security Architect experienced in IEC 62443 and the Purdue model. Think deeply to thoroughly analyse the diagram and provide a structured architectural narration strictly based on visible content.

Figure 7: No-Guardrails architectural reconstruction prompt

You are a senior cyber security expert with over 20 years of experience in cyber-physical systems (CPS) risk and threat modelling, including deep expertise in STRIDE-LM and safety/security co-analysis. You have applied STRIDE-LM extensively in ICS, SCADA, and related CPS domains. Your task is to think deeply to thoroughly analyse the provided system architectural diagram (e.g., Data Flow Diagram) along with any accompanying documentation to produce a comprehensive list of specific threat scenarios relevant to the application.

Figure 8: No-Guardrails threat modelling prompt

### 3.3 Evaluation Metrics

Reconstruction quality is evaluated across four objective, no-ground-truth metrics capturing architectural reconstruction accuracy:

- **Valid Trust Boundaries:** The count of identified security perimeters that respect documented communication protocols and logical isolation (e.g., wireless sensor to gateway transitions, rather than direct EHR to wearable interconnections).
- **Graph Connectivity:** The ratio of network nodes successfully reachable from documented entry points, representing the structural integrity of the reconstructed CPS topology.
- **Impossible Attack Paths:** The count of generated attack paths that violate physical or logical network segmentation (e.g., a direct connection from a field-level inverter to a SCADA system without passing through an intermediary aggregator or gateway).
- **Semantic Coherence:** An *LLM-as-a-judge* [7] metric quantifying the logical consistency and technical clarity of the reconstructed architecture. The scoring prompt directs an independent LLM to act as an impartial judge: *“Please act as an impartial judge and evaluate how semantically coherent is this architecture description for CPS security analysis? Be as objective as possible. Please rate on a Likert scale of 1-5.”*

### 3.4 Metric Acquisition

Metrics were acquired through a two-stage process combining expert visual inspection against reference architectures with automated LLM-based verification.

**Semantic Coherence.** Evaluated via the *LLM-as-a-judge* framework, this metric assesses the logical consistency and technical actionability of the architecture’s markdown output. The prompt directs an independent model to rate the description on a 1–5 Likert scale, quantifying how

effectively the narrative supports downstream security analysis. An example of the judge’s scoring output is provided in Figure 9.

Coherence Rating: 4.9/5.0

Outstanding comprehensive ICS CPS architecture description demonstrating precise Purdue L4→L3.5→L3→L2→L1→L0 zone alignment with DMZ specification, detailed bidirectional IT/OT data flows across fieldbus→OPC UA→REST protocols, comprehensive wireless/safety protocol coverage, and systematic threat modeling matching established IEC 62443 methodology standards.

Figure 9: Semantic coherence evaluation output

**Valid Trust Boundaries.** Boundaries identified within the reconstructed architecture were validated visually against Purdue reference models and case study documentation. This was further confirmed via a targeted LLM prompt: *“Review this architecture narrative and count the number of valid trust boundaries.”*

**Graph Connectivity and Impossible Attack Paths.** Generated attack model markdown was parsed into a directed graph. Connectivity was computed as the fraction of documented entry-point nodes reachable via BFS/DFS from canonical entry points (e.g. internet-facing interfaces). Impossible paths were flagged via visual review against segmentation rules and confirmed by LLM prompt: *“Parse this attack graph markdown. Identify impossible attack paths such as those violating physical or logical network segmentation (e.g. field → SCADA sans aggregator). Compute: reachable nodes from entries / total nodes; impossible paths count. Output format: conn=0.XX, impos=X.”*

An example of the LLM’s output is shown in Figure 10.

conn=0.95, impos=4

Parsed 100+ nodes from ICS CPS attack graph spanning G02-G08 STRIDE goals with Purdue level mappings.

Identified violations: direct wireless sensor→SCADA sans L1 PLC mediation; flat IT/OT network lacking DMZ boundary; maintenance workstation→PLC bypassing L2 supervisory controls; unsegmented historian→field device access path.

Figure 10: Connectivity & impossible path evaluation results

### 3.5 Statistical Analysis

Statistical significance was assessed using the Wilcoxon signed-rank test comparing each variant against the baseline across the three runs ( $n = 3$ ). Ties (identical scores) contribute zero rank; positive ranks (variant  $>$  baseline) sum to  $W_+$ , negative ranks (variant  $<$  baseline) to  $W_-$ , total non-zero ranks  $W = W_+ + W_-$ . The test statistic is  $\min(W_+, W_-)$ ; p-values were computed via exact permutation for small  $n$ .

For example, Trust BD scores baseline: 7,7,6; Text-Only: 5,5,5 yield paired differences, absolute ranks all negative, so  $W_+ = 0$ ,  $W_- = 6$ ,  $W = 6$ ,  $p=0.051$  (one-sided).

The exploratory threshold  $p=0.10$  reflects the family-wise error rate under independent tests, treated as suggestive rather than confirmatory due to small  $n$  and multiple testing. Bold  $p$ -values in tables indicate suggestive degradation.

### 3.6 Experimental Results

This subsection presents the ablation results for the three case studies, comparing the Full ASTRAL baseline against the four variants across all metrics and statistical tests.

Table 2: Ablation Experimental Results: FrostyGoop

| Setting       |          | Run 1 | Run 2 | Run 3 | Average | SD   | W+   | W-   | W    | p-value       |
|---------------|----------|-------|-------|-------|---------|------|------|------|------|---------------|
| Baseline      | Trust BD | 7     | 4     | 4     | 5.10    | 1.29 | –    | –    | –    | –             |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.8   | 4.9   | 4.9   | 4.85    | 0.05 | –    | –    | –    | –             |
| Text-Only     | Trust BD | 4     | 5     | 3     | 3.90    | 0.88 | 31   | 5    | 5    | <b>0.0325</b> |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.4   | 4.6   | 4.3   | 4.58    | 0.14 | 55   | 0    | 0    | <b>0.0023</b> |
| No-Guardrails | Trust BD | 3     | 5     | 6     | 4.00    | 1.05 | 34.5 | 10.5 | 10.5 | <b>0.0754</b> |
|               | Conn     | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | Impos.   | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | Coh.     | 4.9   | 4.8   | 4.9   | 4.89    | 0.03 | 3.5  | 17.5 | 3.5  | <b>0.0512</b> |
| High Temp     | Trust BD | 8     | 8     | 8     | 7.50    | 1.08 | 0    | 55   | 0    | <b>0.0024</b> |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.7   | 4.6   | 4.7   | 4.64    | 0.18 | 48   | 7    | 7    | <b>0.0181</b> |
| Low top- $p$  | Trust BD | 7     | 9     | 6     | 6.70    | 1.42 | 1.5  | 26.5 | 1.5  | <b>0.0165</b> |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.7   | 4.9   | 4.9   | 4.75    | 0.16 | 25   | 3    | 3    | <b>0.0303</b> |

*Legend:* **Trust BD** = valid trust boundaries; **Conn** = graph connectivity; **Impos.** = impossible attack paths; **Coh.** = semantic coherence. Only the first 3 experimental runs are shown for brevity; Average and SD are calculated across all 10 runs.  $W_+$ ,  $W_-$ , and  $W$  denote Wilcoxon signed-rank test statistics.  $p$ -values are reported only for pairs that differ from their baseline values; a **bold**  $p$ -value indicates a result below our pre-specified exploratory threshold of 0.10, which we treat as suggestive rather than confirmatory evidence.

Table 3: Ablation Experimental Results: Medical CPS (MCPS)

| Setting       |          | Run 1 | Run 2 | Run 3 | Average | SD   | W+   | W-   | W    | p-value       |
|---------------|----------|-------|-------|-------|---------|------|------|------|------|---------------|
| Baseline      | Trust BD | 6     | 4     | 6     | 4.80    | 1.32 | –    | –    | –    | –             |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.8   | 4.9   | 4.9   | 4.84    | 0.07 | –    | –    | –    | –             |
| Text-Only     | Trust BD | 3     | 3     | 5     | 4.10    | 1.29 | 31.5 | 13.5 | 13.5 | 0.1400        |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.4   | 4.6   | 4.7   | 4.69    | 0.14 | 28   | 0    | 0    | <b>0.0085</b> |
| No-Guardrails | Trust BD | 7     | 6     | 5     | 4.30    | 1.42 | 24   | 12   | 12   | 0.1942        |
|               | Conn     | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | Impos.   | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | Coh.     | 4.8   | 4.8   | 4.7   | 4.89    | 0.10 | 11.5 | 24.5 | 11.5 | 0.1781        |
| High Temp     | Trust BD | 7     | 8     | 8     | 7.90    | 1.60 | 0    | 45   | 0    | <b>0.0038</b> |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.9   | 4.8   | 4.8   | 4.82    | 0.04 | 11.5 | 24.5 | 11.5 | 0.1781        |
| Low top- $p$  | Trust BD | 10    | 7     | 6     | 8.60    | 1.65 | 0    | 45   | 0    | <b>0.0036</b> |
|               | Conn     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | Impos.   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | Coh.     | 4.9   | 4.9   | 4.7   | 4.83    | 0.08 | 14.5 | 13.5 | 13.5 | 0.4656        |

*Legend:* **Trust BD** = valid trust boundaries; **Conn** = graph connectivity; **Impos.** = impossible attack paths; **Coh.** = semantic coherence. Only the first 3 experimental runs are shown for brevity; Average and SD are calculated across all 10 runs.  $W_+$ ,  $W_-$ , and  $W$  denote Wilcoxon signed-rank test statistics.  $p$ -values are reported only for pairs that differ from their baseline values; a **bold**  $p$ -value indicates a result below our pre-specified exploratory threshold of 0.10, which we treat as suggestive rather than confirmatory evidence.

Table 4: Ablation Experimental Results: Solar PV Inverter

| Setting       |                 | Run 1 | Run 2 | Run 3 | Average | SD   | W+   | W-   | W    | p-value       |
|---------------|-----------------|-------|-------|-------|---------|------|------|------|------|---------------|
| Baseline      | <b>Trust BD</b> | 8     | 8     | 7     | 6.50    | 1.65 | –    | –    | –    | –             |
|               | <b>Conn</b>     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Impos.</b>   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Coh.</b>     | 4.8   | 4.7   | 4.5   | 4.73    | 0.13 | –    | –    | –    | –             |
| Text-Only     | <b>Trust BD</b> | 5     | 5     | 4     | 3.70    | 0.95 | 45   | 0    | 0    | <b>0.0036</b> |
|               | <b>Conn</b>     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Impos.</b>   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Coh.</b>     | 4.7   | 4.7   | 4.3   | 4.58    | 0.18 | 42.5 | 2.5  | 2.5  | <b>0.0083</b> |
| No-Guardrails | <b>Trust BD</b> | 2     | 6     | 3     | 4.70    | 1.49 | 38.5 | 6.5  | 6.5  | <b>0.0284</b> |
|               | <b>Conn</b>     | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | <b>Impos.</b>   | –     | –     | –     | –       | –    | –    | –    | –    | –             |
|               | <b>Coh.</b>     | 4.9   | 4.8   | 4.9   | 4.81    | 0.07 | 11.5 | 33.5 | 11.5 | <b>0.0939</b> |
| High Temp     | <b>Trust BD</b> | 9     | 7     | 7     | 6.70    | 1.77 | 20   | 25   | 20   | 0.3814        |
|               | <b>Conn</b>     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Impos.</b>   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Coh.</b>     | 4.8   | 4.7   | 4.8   | 4.74    | 0.13 | 11.5 | 16.5 | 11.5 | 0.3357        |
| Low top- $p$  | <b>Trust BD</b> | 4     | 5     | 4     | 5.40    | 1.43 | 38   | 17   | 17   | 0.1399        |
|               | <b>Conn</b>     | 1     | 1     | 1     | 1.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Impos.</b>   | 0     | 0     | 0     | 0.00    | 0.00 | –    | –    | –    | –             |
|               | <b>Coh.</b>     | 4.9   | 4.6   | 2.3   | 4.00    | 0.97 | 45.5 | 9.5  | 9.5  | <b>0.0328</b> |

*Legend:* **Trust BD** = valid trust boundaries; **Conn** = graph connectivity; **Impos.** = impossible attack paths; **Coh.** = semantic coherence. Only the first 3 experimental runs are shown for brevity; Average and SD are calculated across all 10 runs.  $W_+$ ,  $W_-$ , and  $W$  denote Wilcoxon signed-rank test statistics.  $p$ -values are reported only for pairs that differ from their baseline values; a **bold**  $p$ -value indicates a result below our pre-specified exploratory threshold of 0.10, which we treat as suggestive rather than confirmatory evidence.

Table 5: Summary: Ablation Study Results Across Three Case Studies (Mean  $\pm$  SD,  $n = 10$ )

| Case Study               | Variant       | Trust BD          | Conn.           | Impos.          | Coherence         |
|--------------------------|---------------|-------------------|-----------------|-----------------|-------------------|
| <b>FrostyGoop</b>        | Full ASTRAL   | $5.10 \pm 1.29$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.85 \pm 0.05$   |
|                          | Text-Only     | $3.90^* \pm 0.88$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.58^* \pm 0.14$ |
|                          | No-Guardrails | $4.00^* \pm 1.05$ | –               | –               | $4.89^* \pm 0.03$ |
|                          | High Temp     | $7.50^* \pm 1.08$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.64^* \pm 0.18$ |
|                          | Low top- $p$  | $6.70^* \pm 1.42$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.75^* \pm 0.16$ |
| <b>Medical CPS</b>       | Full ASTRAL   | $4.80 \pm 1.32$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.84 \pm 0.07$   |
|                          | Text-Only     | $4.10 \pm 1.29$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.69^* \pm 0.14$ |
|                          | No-Guardrails | $4.30 \pm 1.42$   | –               | –               | $4.89 \pm 0.10$   |
|                          | High Temp     | $7.90^* \pm 1.60$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.82 \pm 0.04$   |
|                          | Low top- $p$  | $8.60^* \pm 1.65$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.83 \pm 0.08$   |
| <b>Solar PV Inverter</b> | Full ASTRAL   | $6.50 \pm 1.65$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.73 \pm 0.13$   |
|                          | Text-Only     | $3.70^* \pm 0.95$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.58^* \pm 0.18$ |
|                          | No-Guardrails | $4.70^* \pm 1.49$ | –               | –               | $4.81^* \pm 0.07$ |
|                          | High Temp     | $6.70 \pm 1.77$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.74 \pm 0.13$   |
|                          | Low top- $p$  | $5.40 \pm 1.43$   | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $4.00^* \pm 0.97$ |

*Legend:* **Trust BD** = valid trust boundaries; **Conn.** = graph connectivity; **Impos.** = impossible attack paths; **Coherence** = semantic coherence (scored 1–5). Results are reported as Mean  $\pm$  SD across  $n = 10$  experimental runs. The \* denotes statistical significance at our exploratory threshold ( $p \leq 0.10$ ) using the Wilcoxon signed-rank test against the Full ASTRAL baseline.

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